In [1]:
```
%matplotlib ipympl
import pandas as pd
import numpy as np
import matplotlib.pylab as plt
from scipy import stats
from matplotlib.backends.backend_pdf import PdfPages
import seaborn as sns
```

Seaborn and Correlation Matrix

Today we are going to load seaborn and make on of their plots. They have some nice plots and it is becoming mainstream. https://seaborn.pydata.org/examples/index.html

First see if you have it. Type import seaborn as sns

If not to install there are two ways. First I am going to try the new way.

- Open Anaconda Navigator
- goto environments
- where is says installed click not installed or all
- search for seaborn and check it.

If that doesn't work we can do it in the terminal.

- open a new terminal window
- googel conda install seaborn
- open the page
- copy the "conda install -c anaconda seaborn"
- paste into temrinal and run

If those all fail find me.

Read in our well data

In [2]:
```
df=pd.read_csv('well_data.csv')
```

List all the columns

In [3]:
```
df.columns
```

Out[3]:
```
Index(['Well_ID', 'Lat', 'Lon', 'Depth', 'Drink', 'Si', 'P', 'S', 'Ca', 'Fe',
       'Ba', 'Na', 'Mg', 'K', 'Mn', 'As', 'Sr', 'F', 'Cl', 'SO4', 'Br'],
      dtype='object')
```

If you ever need to delete a column this is how you do it. (not needed)

In [4]:
```
df = df.drop(['Well_ID','Drink'], axis=1)
```

Go back and find the columns you want to use and save it back to the dataframe.

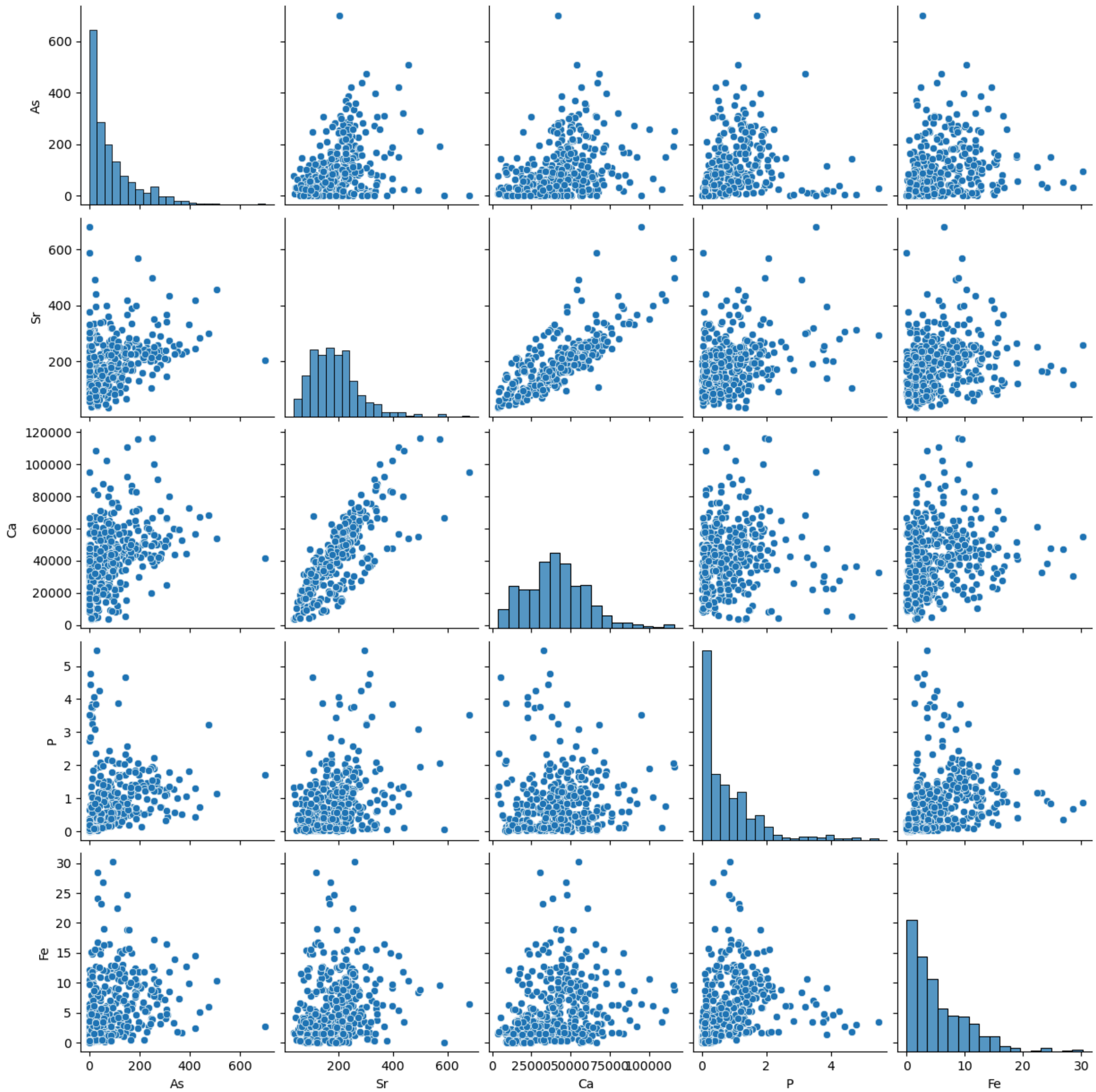
In [5]:
```
df=df[['As', 'Sr', 'Ca', 'P', 'Fe']]
```

Run seaborn pairplot. https://seaborn.pydata.org/generated/seaborn.pairplot.html

In [6]:
```
sns.pairplot(df)
plt.savefig('test.jpg')
```

```
/Users/bmaillou/anaconda3/envs/BigDataPython2025/lib/python3.8/site-packages/seaborn/axisgrid.py:12
3: UserWarning: The figure layout has changed to tight
  self._figure.tight_layout(*args, **kwargs)
```

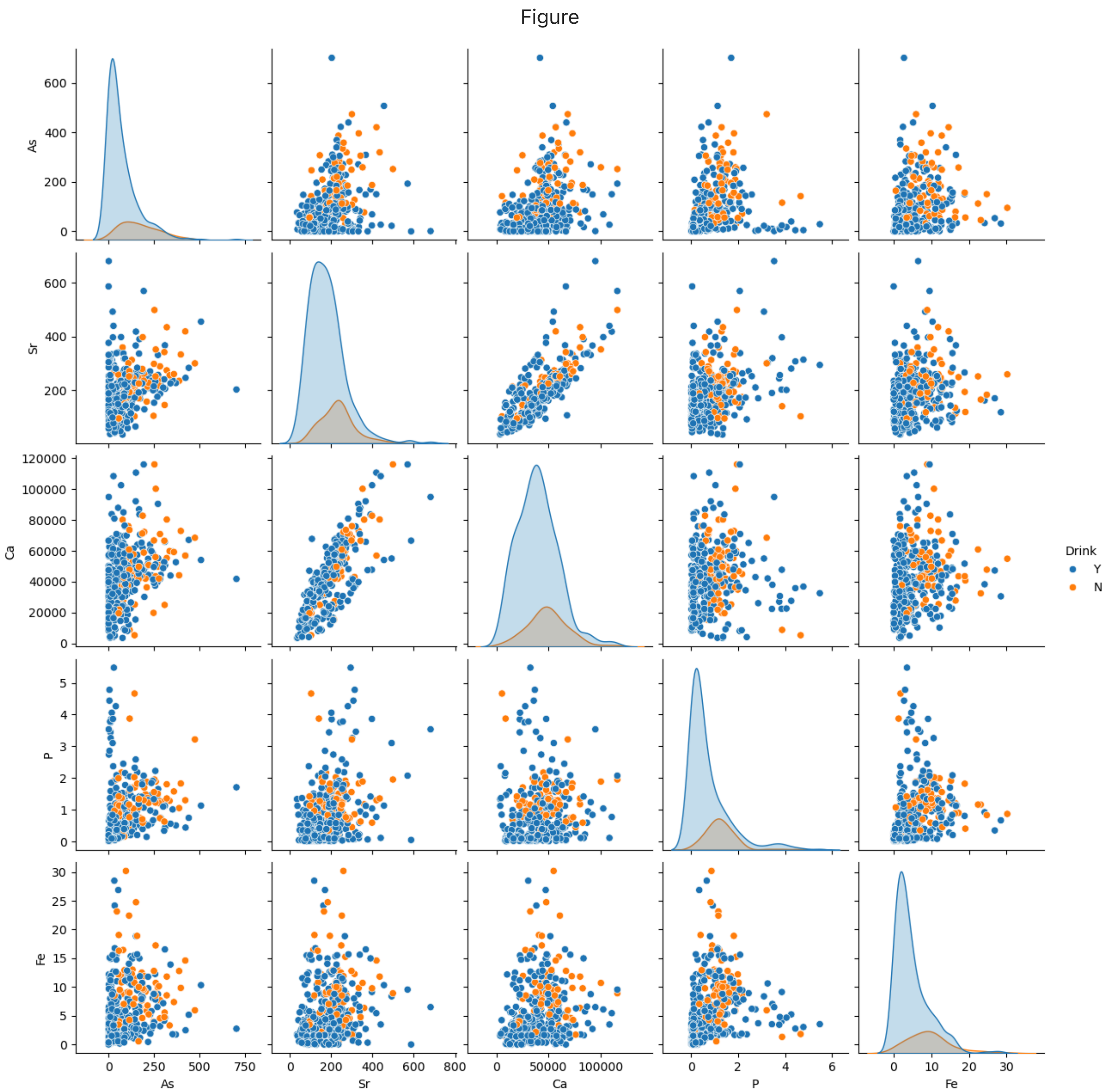
Figure



Also you can color by a categorical value. So add 'Drink' back in and use it to set hue

In [7]:
```
df=pd.read_csv('well_data.csv')
df=df[['As', 'Sr', 'Ca', 'P', 'Fe','Drink']]
sns.pairplot(df,hue='Drink')
plt.savefig('test.jpg')
```

```
/Users/bmaillou/anaconda3/envs/BigDataPython2025/lib/python3.8/site-packages/seaborn/axisgrid.py:12
3: UserWarning: The figure layout has changed to tight
  self._figure.tight_layout(*args, **kwargs)
```



If it is easier you could do it by column number instead

In [8]:
```
df=pd.read_csv('well_data.csv')
```

In [9]:
```
for count,col in enumerate(df):
    print(count,col)
```

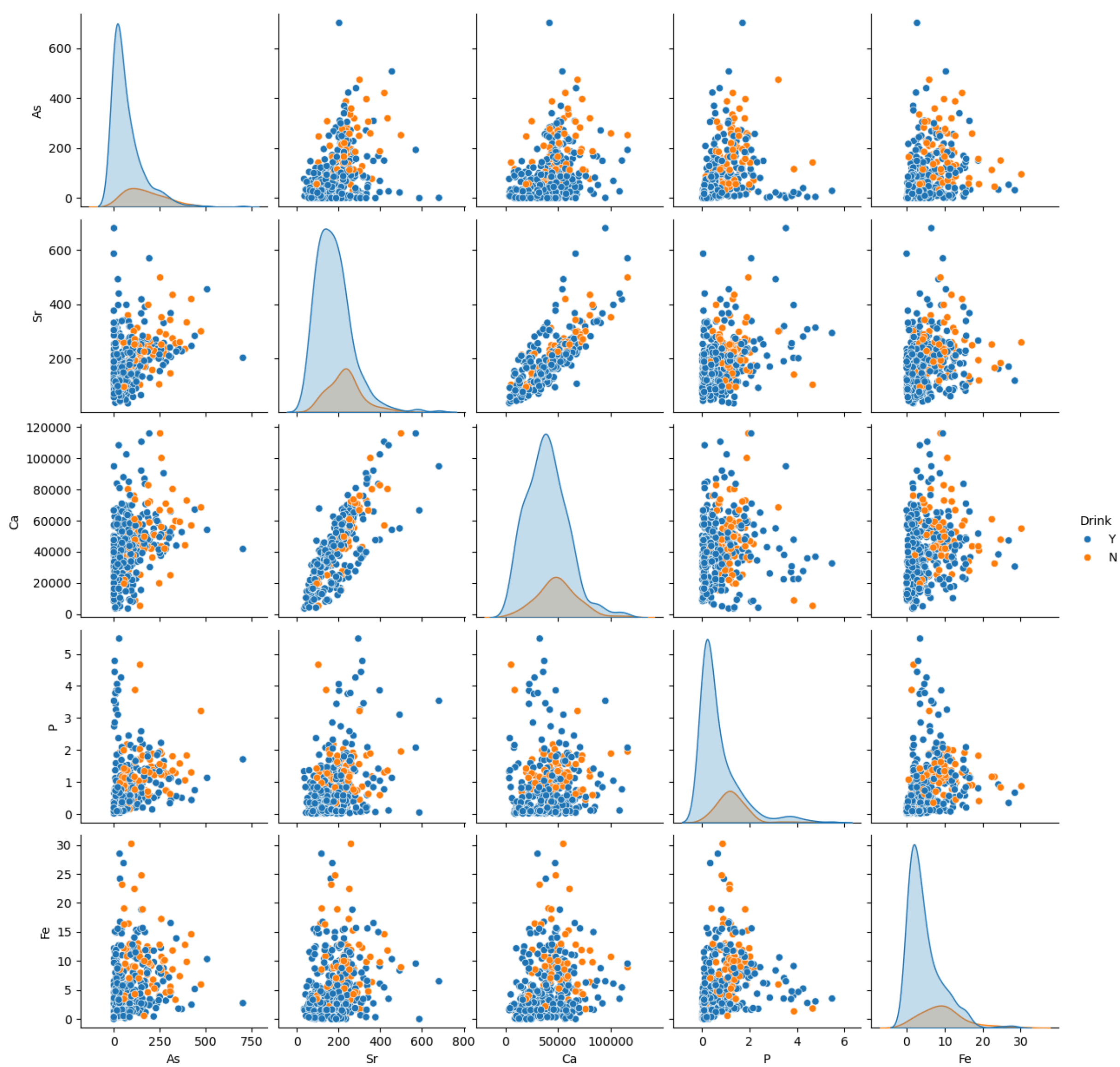
```
0 Well_ID
1 Lat
2 Lon
3 Depth
4 Drink
5 Si
6 P
7 S
8 Ca
9 Fe
10 Ba
11 Na
12 Mg
13 K
14 Mn
15 As
16 Sr
17 F
18 Cl
19 SO4
20 Br
```

In [10]:

```
df=df.iloc[:,[15, 16, 8, 6, 9,4]]
sns.pairplot(df,hue='Drink')
plt.savefig('test.jpg')
```

```
/Users/bmaillou/anaconda3/envs/BigDataPython2025/lib/python3.8/site-packages/seaborn/axisgrid.py:12
3: UserWarning: The figure layout has changed to tight
  self._figure.tight_layout(*args, **kwargs)
```

Figure




```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

This is all extra material below that I found fun

But you don't get all the r-values or p-values. Can you make an excel file of all of them? I would

1. Just take all the numerical data. See first line for how I did it.
2. Make a double for loop to go through the data
3. Calculate the r2
4. print to your new dataframe
5. save to excel

In [16]:
```
df=df_well_data.select_dtypes(include=np.number)
df_r2=pd.DataFrame()

for i in df:
```

```
    for j in df:
        print (i,j)
        results=stats.linregress(df_well_data[[i,j]].dropna())
        df_r2.at[j,i]=results.rvalue**2

df_r2.to_excel('All_correlations.xlsx')
```

```
Well_ID Well_ID
Well_ID Lat
Well_ID Lon
Well_ID Depth
Well_ID Si
Well_ID P
Well_ID S
Well_ID Ca
Well_ID Fe
Well_ID Ba
Well_ID Na
Well_ID Mg
Well_ID K
Well_ID Mn
Well_ID As
Well_ID Sr
Well_ID F
Well_ID Cl
Well_ID SO4
Well_ID Br
Lat Well_ID
Lat Lat
Lat Lon
Lat Depth
Lat Si
Lat P
Lat S
Lat Ca
Lat Fe
Lat Ba
Lat Na
Lat Mg
Lat K
Lat Mn
Lat As
Lat Sr
Lat F
Lat Cl
Lat SO4
Lat Br
Lon Well_ID
Lon Lat
Lon Lon
Lon Depth
Lon Si
Lon P
Lon S
Lon Ca
Lon Fe
Lon Ba
Lon Na
Lon Mg
Lon K
Lon Mn
Lon As
Lon Sr
Lon F
Lon Cl
Lon SO4
Lon Br
Depth Well_ID
Depth Lat
Depth Lon
Depth Depth
Depth Si
Depth P
Depth S
Depth Ca
Depth Fe
Depth Ba
```

```
Depth Na
Depth Mg
Depth K
Depth Mn
Depth As
Depth Sr
Depth F
Depth Cl
Depth SO4
Depth Br
Si Well_ID
Si Lat
Si Lon
Si Depth
Si Si
Si P
Si S
Si Ca
Si Fe
Si Ba
Si Na
Si Mg
Si K
Si Mn
Si As
Si Sr
Si F
Si Cl
Si SO4
Si Br
P Well_ID
P Lat
P Lon
P Depth
P Si
P P
P S
P Ca
P Fe
P Ba
P Na
P Mg
P K
P Mn
P As
P Sr
P F
P Cl
P SO4
P Br
S Well_ID
S Lat
S Lon
S Depth
S Si
S P
S S
S Ca
S Fe
S Ba
S Na
S Mg
S K
S Mn
S As
S Sr
S F
S Cl
S SO4
S Br
```

```
Ca Well_ID
Ca Lat
Ca Lon
Ca Depth
Ca Si
Ca P
Ca S
Ca Ca
Ca Fe
Ca Ba
Ca Na
Ca Mg
Ca K
Ca Mn
Ca As
Ca Sr
Ca F
Ca Cl
Ca SO4
Ca Br
Fe Well_ID
Fe Lat
Fe Lon
Fe Depth
Fe Si
Fe P
Fe S
Fe Ca
Fe Fe
Fe Ba
Fe Na
Fe Mg
Fe K
Fe Mn
Fe As
Fe Sr
Fe F
Fe Cl
Fe SO4
Fe Br
Ba Well_ID
Ba Lat
Ba Lon
Ba Depth
Ba Si
Ba P
Ba S
Ba Ca
Ba Fe
Ba Ba
Ba Na
Ba Mg
Ba K
Ba Mn
Ba As
Ba Sr
Ba F
Ba Cl
Ba SO4
Ba Br
Na Well_ID
Na Lat
Na Lon
Na Depth
Na Si
Na P
Na S
Na Ca
Na Fe
Na Ba
```

```
Na Na
Na Mg
Na K
Na Mn
Na As
Na Sr
Na F
Na Cl
Na SO4
Na Br
Mg Well_ID
Mg Lat
Mg Lon
Mg Depth
Mg Si
Mg P
Mg S
Mg Ca
Mg Fe
Mg Ba
Mg Na
Mg Mg
Mg K
Mg Mn
Mg As
Mg Sr
Mg F
Mg Cl
Mg SO4
Mg Br
K Well_ID
K Lat
K Lon
K Depth
K Si
K P
K S
K Ca
K Fe
K Ba
K Na
K Mg
K K
K Mn
K As
K Sr
K F
K Cl
K SO4
K Br
Mn Well_ID
Mn Lat
Mn Lon
Mn Depth
Mn Si
Mn P
Mn S
Mn Ca
Mn Fe
Mn Ba
Mn Na
Mn Mg
Mn K
Mn Mn
Mn As
Mn Sr
Mn F
Mn Cl
Mn SO4
Mn Br
```

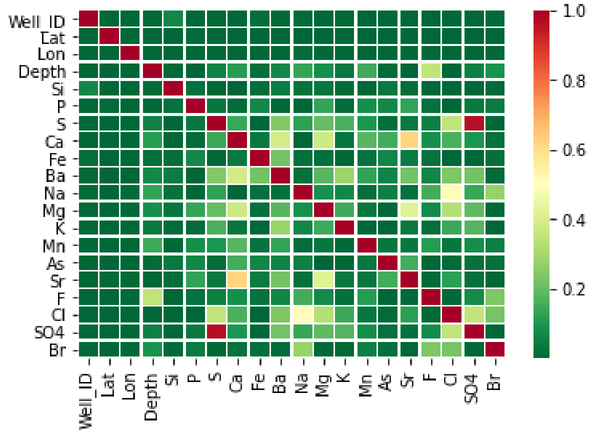
```
As Well_ID
As Lat
As Lon
As Depth
As Si
As P
As S
As Ca
As Fe
As Ba
As Na
As Mg
As K
As Mn
As As
As Sr
As F
As Cl
As SO4
As Br
Sr Well_ID
Sr Lat
Sr Lon
Sr Depth
Sr Si
Sr P
Sr S
Sr Ca
Sr Fe
Sr Ba
Sr Na
Sr Mg
Sr K
Sr Mn
Sr As
Sr Sr
Sr F
Sr Cl
Sr SO4
Sr Br
F Well_ID
F Lat
F Lon
F Depth
F Si
F P
F S
F Ca
F Fe
F Ba
F Na
F Mg
F K
F Mn
F As
F Sr
F F
F Cl
F SO4
F Br
Cl Well_ID
Cl Lat
Cl Lon
Cl Depth
Cl Si
Cl P
Cl S
Cl Ca
Cl Fe
Cl Ba
```

```
Cl Na
Cl Mg
Cl K
Cl Mn
Cl As
Cl Sr
Cl F
Cl Cl
Cl SO4
Cl Br
SO4 Well_ID
SO4 Lat
SO4 Lon
SO4 Depth
SO4 Si
SO4 P
SO4 S
SO4 Ca
SO4 Fe
SO4 Ba
SO4 Na
SO4 Mg
SO4 K
SO4 Mn
SO4 As
SO4 Sr
SO4 F
SO4 Cl
SO4 SO4
SO4 Br
Br Well_ID
Br Lat
Br Lon
Br Depth
Br Si
Br P
Br S
Br Ca
Br Fe
Br Ba
Br Na
Br Mg
Br K
Br Mn
Br As
Br Sr
Br F
Br Cl
Br SO4
Br Br
```

You can open your excel file or visualize in sseaborn.

In [19]:
```
sns.heatmap(df_r2, cmap='RdYlGn_r', linewidths=0.5, annot=False)
```

Out[19]:
```
<AxesSubplot:>
```



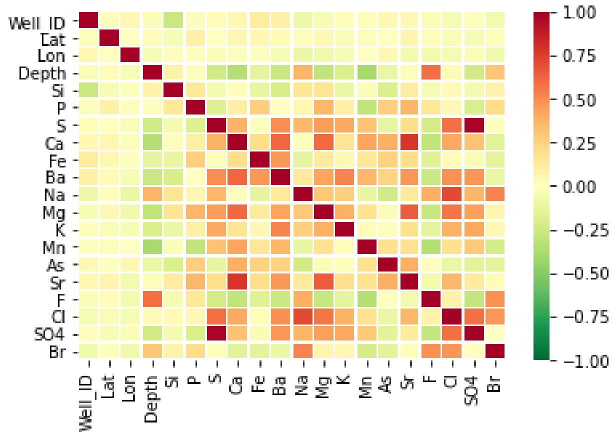
Try with r-vlaues

In [22]:
```
df=df_well_data.select_dtypes(include=np.number)
df_r2=pd.DataFrame()

for i in df:
    for j in df:
        results=stats.linregress(df_well_data[[i,j]].dropna())
        df_r2.at[j,i]=results.rvalue

sns.heatmap(df_r2, cmap='RdYlGn_r', linewidths=0.5, annot=False,vmin=-1,vmax=1)
```

Out[22]: <AxesSubplot:>

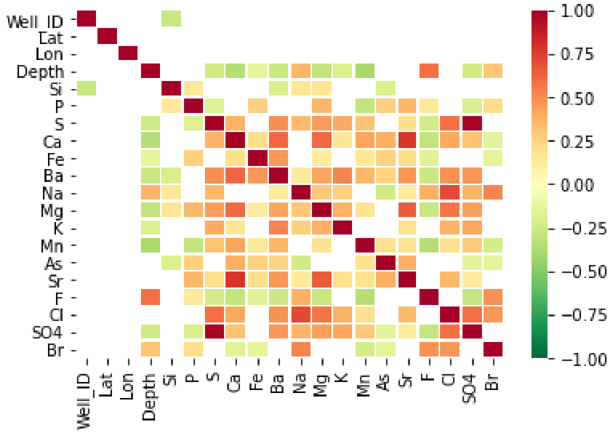


For excel I would do r-values and only save for when p-value is less than 0.01

In [25]:
```
df=df_well_data.select_dtypes(include=np.number)
df_r2=pd.DataFrame()

for i in df:
    for j in df:
        results=stats.linregress(df_well_data[[i,j]].dropna())
        if results.pvalue<0.01:
            df_r2.at[j,i]=results.rvalue
        else:
            df_r2.at[j,i]=np.nan
df_r2.to_excel('r-values.xlsx')
sns.heatmap(df_r2, cmap='RdYlGn_r', linewidths=0.5, annot=False,vmin=-1,vmax=1)
```

Out[25]: <AxesSubplot:>



If column names are really long remember you can use iloc to chosse a datafram by column number

In [29]:
```
df.iloc[:,3:7]
```

Out[29]:

	Depth	Si	P	S
0	45	NaN	NaN	NaN
1	60	NaN	NaN	NaN
2	60	NaN	NaN	NaN
3	50	48084.33842	0.936358	2085.570979
4	150	NaN	NaN	NaN
...	...	...	...	...
754	160	32379.64000	0.197380	3669.430000
755	60	25561.12000	0.090570	13771.370000
756	45	31319.48000	1.162550	38.300000
757	60	30605.53000	1.556120	4168.520000
758	50	NaN	NaN	NaN

759 rows × 4 columns

In [ ]: